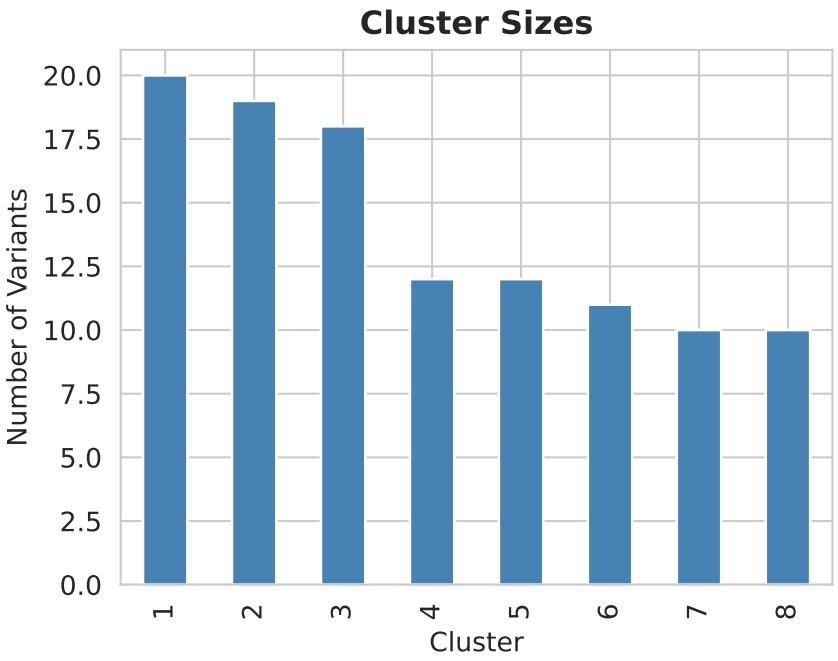
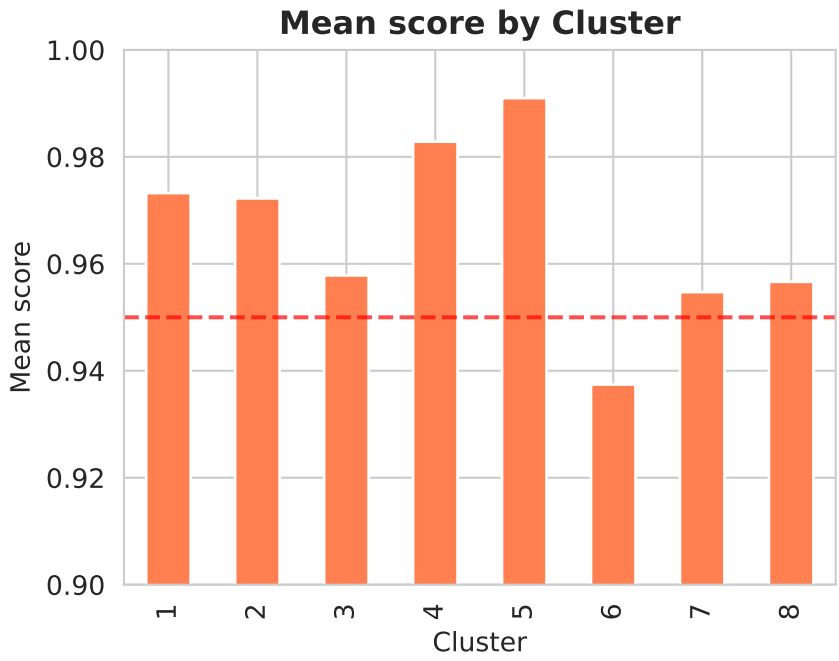
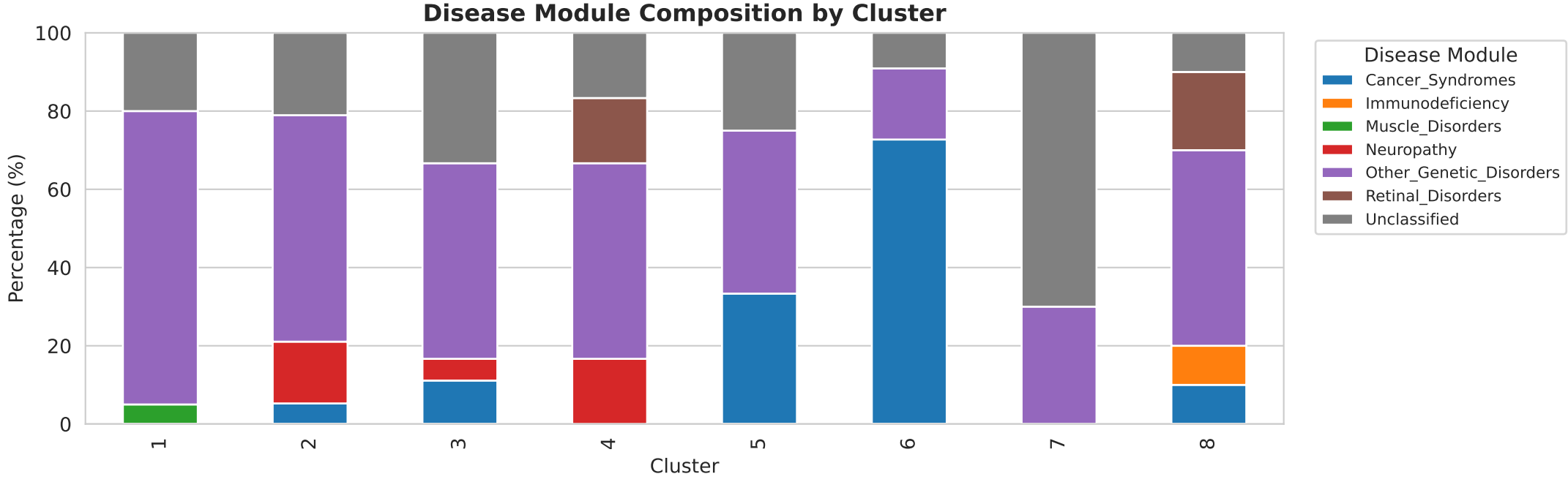
Comprehensive Disease Biology Analysis - Overview







Gene and Disease Module Patterns



Cluster 1: Comprehensive Analysis

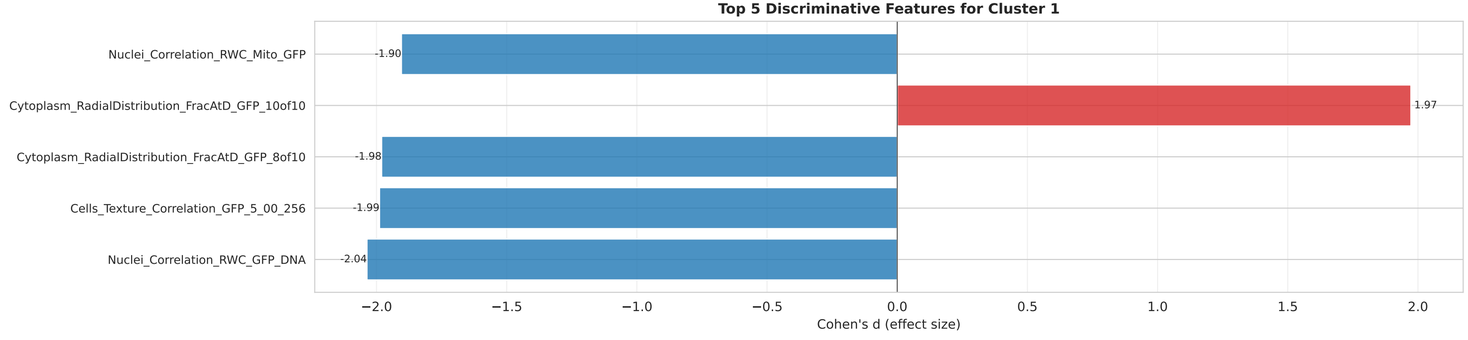
```
CLUSTER 1 ANALYSIS
================================================================================================

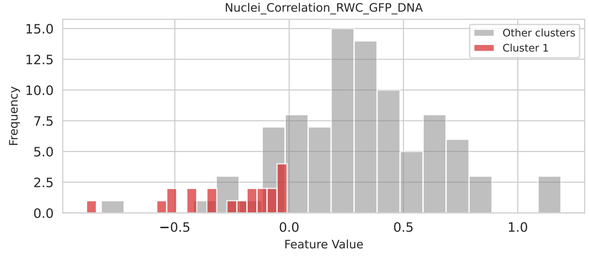
Size: 20 variants
Mean score: 0.973
Hit variants: 20 (100.0%)

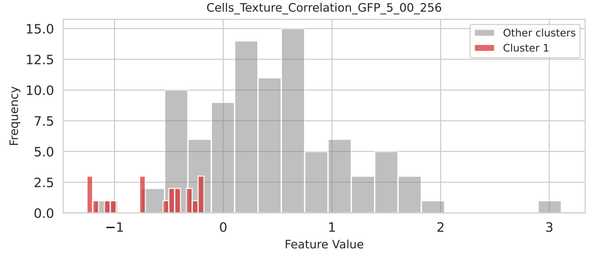
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BFSP2_Arg287Trp         |  BFSP2_Arg339His         |  DES_Tyr122Asp           |  DES_Arg127Pro
  GFAP_Asn77Ser           |  GFAP_Arg79Cys           |  GFAP_Arg79Ser           |  GFAP_Arg88Cys
  GFAP_Leu90Pro           |  GFAP_Leu235Pro          |  GFAP_Lys236Thr          |  GFAP_Arg239Leu
  GFAP_Arg239Pro          |  GFAP_Lys279Glu          |  GFAP_Leu352Pro          |  GFAP_Tyr366Cys
  GFAP_Glu371Gly          |  KRT8_Tyr54His           |  KRT8_Ile63Val           |  KRT8_Gly434Ser

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 15 [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, DES_Tyr122Asp, DES_Arg127Pro, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  Rare genetic eye disease: 13 [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  Rare genetic cardiac disease: 2 [DES_Tyr122Asp, DES_Arg127Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: ALEXANDER DISEASE (12x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  Disease: HEPATITIS C VIRUS, SUSCEPTIBILITY TO (2x) [KRT8_Tyr54His, KRT8_Ile63Val]
  inheritance_pattern: autosomal dominant (13x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, BFSP2_Arg287Trp, GFAP_Arg79Cys, GFAP_Leu352Pro, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  inheritance_pattern: unknown (6x) [DES_Tyr122Asp, KRT8_Gly434Ser, KRT8_Tyr54His, KRT8_Ile63Val, GFAP_Arg79Ser, BFSP2_Arg339His]
  pLDDT_Category: >90 (13x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, BFSP2_Arg287Trp, GFAP_Arg79Cys, GFAP_Leu352Pro, DES_Tyr122Asp, DES_Arg127Pro, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Lys279Glu]
  pLDDT_Category: ≤50 (3x) [KRT8_Gly434Ser, KRT8_Tyr54His, KRT8_Ile63Val]
  pLDDT_Category: 70-90 (3x) [GFAP_Leu235Pro, GFAP_Glu371Gly, GFAP_Lys236Thr]
  RSA_Category: 10-90 (17x) [GFAP_Arg239Leu, GFAP_Arg88Cys, GFAP_Asn77Ser, BFSP2_Arg287Trp, GFAP_Arg79Cys, GFAP_Leu352Pro, DES_Tyr122Asp, KRT8_Gly434Ser, DES_Arg127Pro, GFAP_Arg79Ser, GFAP_Leu90Pro, GFAP_Arg239Pro, GFAP_Tyr366Cys, GFAP_Leu235Pro, GFAP_Lys279Glu, GFAP_Glu371Gly, GFAP_Lys236Thr]
  RSA_Category: ⩾90 (2x) [KRT8_Tyr54His, KRT8_Ile63Val]
```







Cluster 2: Comprehensive Analysis

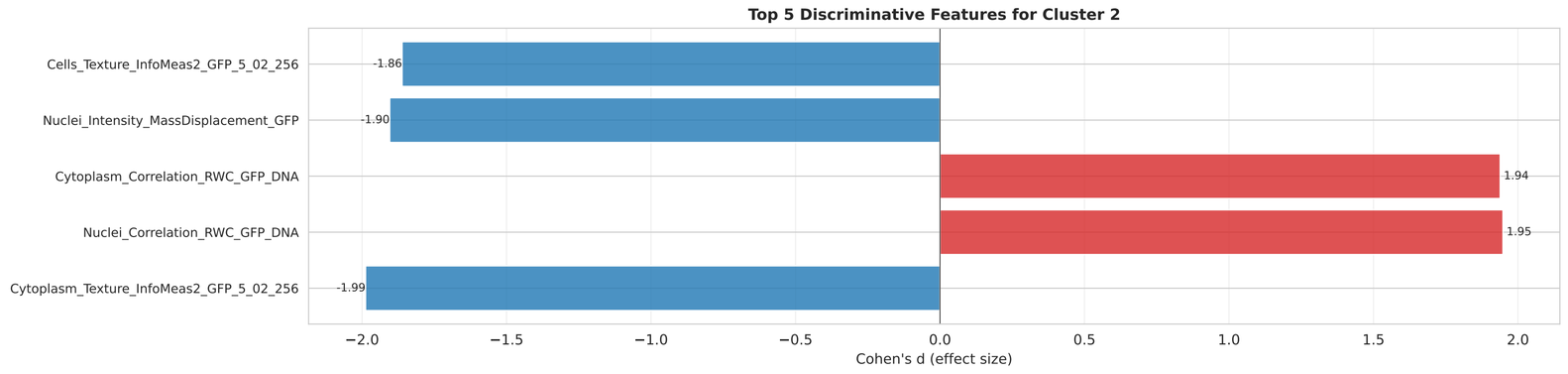
```
CLUSTER 2 ANALYSIS
================================================================================================

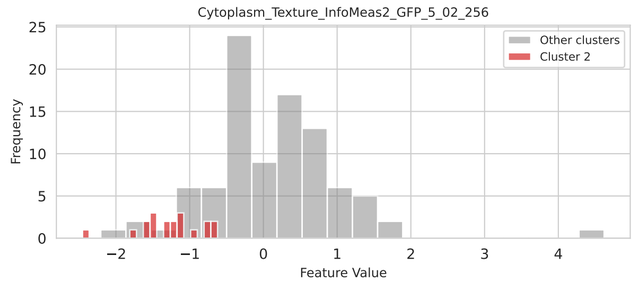
Size: 19 variants
Mean score: 0.972
Hit variants: 19 (100.0%)

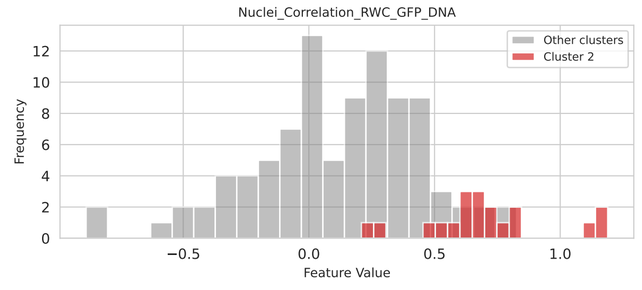
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  BRCA1_Glu455Lys         |  FXN_Met1Ile            |  HPRT1_Phe74Leu         |  IL36RN_Leu27Pro
  KRT4_Glu449Lys          |  LITAF_Trp116Gly        |  LITAF_Ser118Pro        |  LITAF_Leu122Val
  LITAF_Phe136Leu         |  LITAF_Val144Met        |  MVK_Leu41Pro           |  MVK_Leu255Pro
  PLP1_Gly74Arg           |  PLP1_Trp128Arg         |  PNPO_Arg95Cys          |  PNPO_Arg95His
  PNPO_Arg116Gln          |  SMARCB1_Arg37His       |  SMARCB1_Pro48Leu

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 14 [FXN_Met1Ile, LITAF_Trp116Gly, LITAF_Leu122Val, PNPO_Arg95Cys, PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu, LITAF_Ser118Pro, LITAF_Val144Met, LITAF_Phe136Leu, PLP1_Gly74Arg, HPRT1_Phe74Leu]
  Rare inborn errors of metabolism: 6 [MVK_Leu41Pro, MVK_Leu255Pro, PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln, HPRT1_Phe74Leu]
  Rare genetic skin disease: 6 [MVK_Leu41Pro, IL36RN_Leu27Pro, MVK_Leu255Pro, BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic eye disease: 5 [FXN_Met1Ile, MVK_Leu41Pro, MVK_Leu255Pro, PLP1_Trp128Arg, PLP1_Gly74Arg]
  Rare genetic developmental defect during embryogenesis: 5 [MVK_Leu41Pro, MVK_Leu255Pro, BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic tumor: 4 [KRT4_Glu449Lys, BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic immune disease: 3 [MVK_Leu41Pro, IL36RN_Leu27Pro, MVK_Leu255Pro]
  Rare genetic bone disease: 3 [BRCA1_Glu455Lys, SMARCB1_Arg37His, SMARCB1_Pro48Leu]
  Rare genetic systemic or rheumatologic disease: 2 [MVK_Leu41Pro, MVK_Leu255Pro]
  Rare genetic renal disease: 2 [BRCA1_Glu455Lys, HPRT1_Phe74Leu]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: CHARCOT-MARIE-TOOTH DISEASE, DEMYELINATING, TYPE 1C (3x) [LITAF_Trp116Gly, LITAF_Leu122Val, LITAF_Val144Met]
  Disease: PYRIDOXAMINE 5-PRIME-PHOSPHATE OXIDASE DEFICIENCY (3x) [PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln]
  Disease: POROKERATOSIS 3, MULTIPLE TYPES (2x) [MVK_Leu41Pro, MVK_Leu255Pro]
  inheritance_pattern: autosomal dominant (8x) [MVK_Leu41Pro, KRT4_Glu449Lys, LITAF_Trp116Gly, LITAF_Leu122Val, MVK_Leu255Pro, SMARCB1_Arg37His, SMARCB1_Pro48Leu, LITAF_Val144Met]
  inheritance_pattern: autosomal recessive (5x) [FXN_Met1Ile, IL36RN_Leu27Pro, PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln]
  inheritance_pattern: unknown (4x) [LITAF_Ser118Pro, LITAF_Phe136Leu, PLP1_Gly74Arg, HPRT1_Phe74Leu]
  pLDDT_Category: >90 (11x) [MVK_Leu41Pro, KRT4_Glu449Lys, IL36RN_Leu27Pro, MVK_Leu255Pro, PNPO_Arg95Cys, PNPO_Arg95His, PLP1_Trp128Arg, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu, LITAF_Val144Met]
  pLDDT_Category: 70-90 (4x) [LITAF_Trp116Gly, LITAF_Leu122Val, LITAF_Ser118Pro, LITAF_Phe136Leu]
  pLDDT_Category: ≤50 (2x) [FXN_Met1Ile, BRCA1_Glu455Lys]
  RSA_Category: 10-90 (12x) [KRT4_Glu449Lys, LITAF_Trp116Gly, LITAF_Leu122Val, MVK_Leu255Pro, BRCA1_Glu455Lys, PNPO_Arg95Cys, PNPO_Arg95His, PNPO_Arg116Gln, SMARCB1_Arg37His, SMARCB1_Pro48Leu, LITAF_Ser118Pro, LITAF_Phe136Leu]
  RSA_Category: <10 (4x) [MVK_Leu41Pro, IL36RN_Leu27Pro, PLP1_Trp128Arg, LITAF_Val144Met]
```







Cluster 3: Comprehensive Analysis

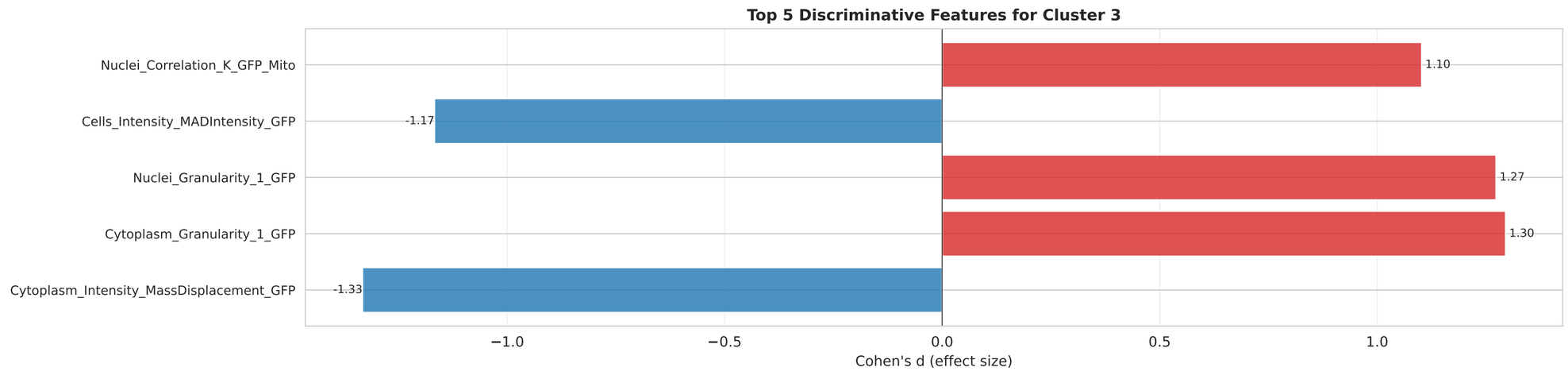
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CLUSTER 3 ANALYSIS
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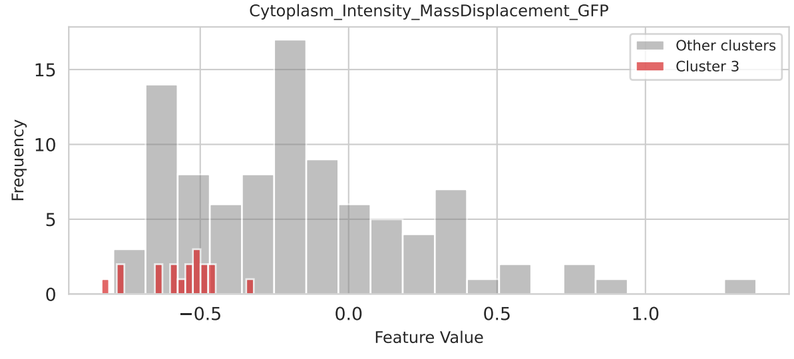
Size: 18 variants
Mean score: 0.958
Hit variants: 18 (100.0%)

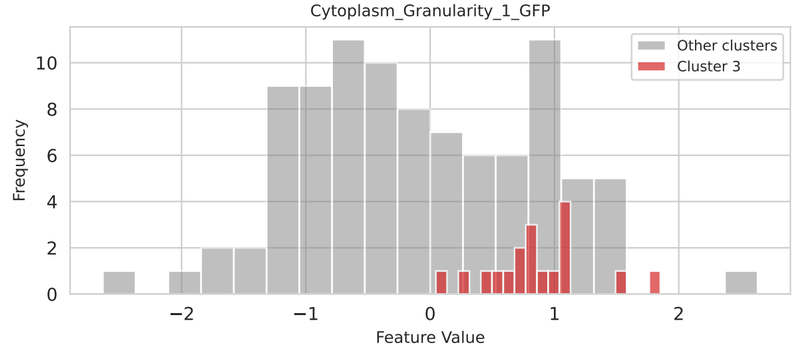
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AGXT_Gly116Arg          |  AGXT_Gly161Arg          |  AGXT_Ile202Asn          |  BRCA1_Met1Ile
  CDKN1A_Asp149Gly        |  FOXP3_Leu242Pro         |  LITAF_Ile92Val          |  MLH1_Ser295Thr
  MLH1_Asp485Glu          |  MLH1_Leu653Pro          |  NT5C3A_Cys108Arg        |  NT5C3A_Asp132Val
  NT5C3A_Gly202Arg        |  NT5C3A_Gly275Arg        |  PYGL_Asn632Ile          |  SFTPA2_Gly231Val
  TTPA_Glu141Lys          |  TTPA_Gly246Arg

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare inborn errors of metabolism: 10 [AGXT_Gly161Arg, AGXT_Ile202Asn, NT5C3A_Gly275Arg, AGXT_Gly116Arg, NT5C3A_Asp132Val, TTPA_Glu141Lys, PYGL_Asn632Ile, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg, TTPA_Gly246Arg]
  Rare genetic eye disease: 5 [AGXT_Gly161Arg, AGXT_Ile202Asn, AGXT_Gly116Arg, TTPA_Glu141Lys, TTPA_Gly246Arg]
  Rare genetic hematologic disease: 5 [NT5C3A_Gly275Arg, BRCA1_Met1Ile, NT5C3A_Asp132Val, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg]
  Rare genetic tumor: 5 [BRCA1_Met1Ile, MLH1_Leu653Pro, MLH1_Ser295Thr, MLH1_Asp485Glu, CDKN1A_Asp149Gly]
  Inherited cancer-predisposing syndrome: 5 [BRCA1_Met1Ile, MLH1_Leu653Pro, MLH1_Ser295Thr, MLH1_Asp485Glu, CDKN1A_Asp149Gly]
  Rare genetic renal disease: 4 [AGXT_Gly161Arg, AGXT_Ile202Asn, BRCA1_Met1Ile, AGXT_Gly116Arg]
  Rare genetic immune disease: 4 [MLH1_Leu653Pro, MLH1_Ser295Thr, MLH1_Asp485Glu, FOXP3_Leu242Pro]
  Rare genetic neurological disorder: 3 [LITAF_Ile92Val, TTPA_Glu141Lys, TTPA_Gly246Arg]
  Rare genetic gastroenterological disease: 2 [BRCA1_Met1Ile, FOXP3_Leu242Pro]
  Rare genetic endocrine disease: 2 [CDKN1A_Asp149Gly, FOXP3_Leu242Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: HYPEROXALURIA, PRIMARY, TYPE I (3x) [AGXT_Gly161Arg, AGXT_Ile202Asn, AGXT_Gly116Arg]
  Disease: ANEMIA, CONGENITAL, NONSPHEROCYTIC HEMOLYTIC, 8 (2x) [NT5C3A_Gly275Arg, NT5C3A_Asp132Val]
  Disease: ATAXIA WITH VITAMIN E DEFICIENCY (2x) [TTPA_Glu141Lys, TTPA_Gly246Arg]
  inheritance_pattern: autosomal recessive (8x) [AGXT_Gly161Arg, AGXT_Ile202Asn, NT5C3A_Gly275Arg, AGXT_Gly116Arg, NT5C3A_Asp132Val, TTPA_Glu141Lys, PYGL_Asn632Ile, TTPA_Gly246Arg]
  inheritance_pattern: unknown (6x) [MLH1_Leu653Pro, MLH1_Asp485Glu, CDKN1A_Asp149Gly, NT5C3A_Cys108Arg, NT5C3A_Gly202Arg, FOXP3_Leu242Pro]
  inheritance_pattern: autosomal dominant (3x) [LITAF_Ile92Val, MLH1_Ser295Thr, SFTPA2_Gly231Val]
  pLDDT_Category: >90 (10x) [AGXT_Gly161Arg, AGXT_Ile202Asn, AGXT_Gly116Arg, MLH1_Leu653Pro, LITAF_Ile92Val, TTPA_Glu141Lys, MLH1_Ser295Thr, PYGL_Asn632Ile, TTPA_Gly246Arg, SFTPA2_Gly231Val]
  RSA_Category: <10 (7x) [AGXT_Gly161Arg, AGXT_Ile202Asn, AGXT_Gly116Arg, MLH1_Leu653Pro, TTPA_Glu141Lys, MLH1_Ser295Thr, SFTPA2_Gly231Val]
  RSA_Category: 10-90 (3x) [LITAF_Ile92Val, PYGL_Asn632Ile, TTPA_Gly246Arg]
```







Cluster 4: Comprehensive Analysis

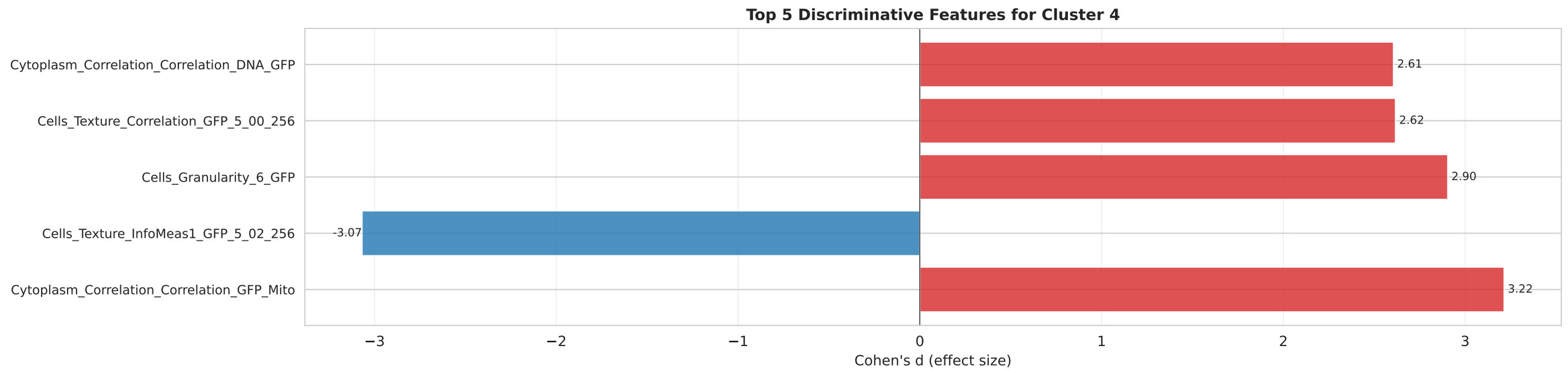
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CLUSTER 4 ANALYSIS
================================================================================================================

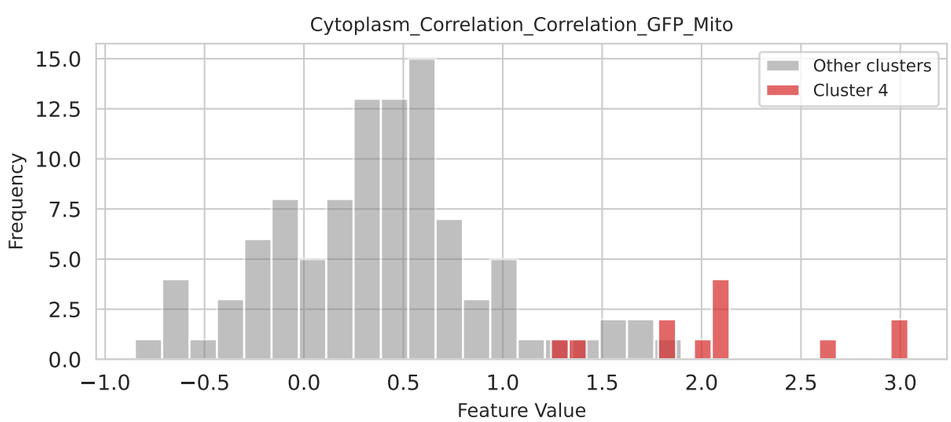
Size: 12 variants
Mean score: 0.983
Hit variants: 12 (100.0%)

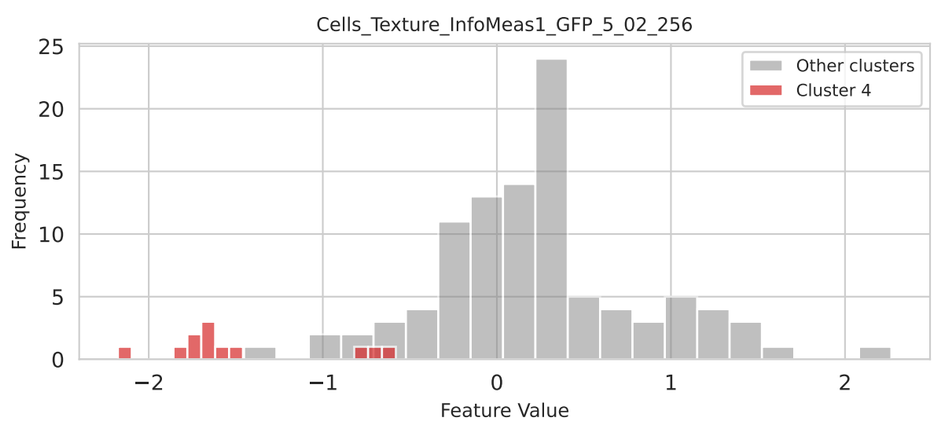
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  PLP1_Thr43Ile           |  PLP1_Thr147Pro          |  PLP1_Ile152Thr          |  PLP1_Gly186Cys
  PLP1_Leu189Pro          |  PLP1_Cys193Tyr          |  PMP22_Leu16Pro          |  PMP22_Leu80Pro
  RHO_Cys110Tyr           |  RHO_Asp190Gly           |  SFTPC_Leu188Gln         |  SFTPC_Leu194Pro

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic neurological disorder: 8 [PMP22_Leu16Pro, PLP1_Thr147Pro, PLP1_Gly186Cys, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Cys193Tyr, PLP1_Leu189Pro, PMP22_Leu80Pro]
  Rare genetic eye disease: 8 [PLP1_Thr147Pro, PLP1_Gly186Cys, RHO_Cys110Tyr, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Cys193Tyr, RHO_Asp190Gly, PLP1_Leu189Pro]
  Genetic otorhinolaryngologic disease: 2 [PMP22_Leu16Pro, PMP22_Leu80Pro]
  Rare chromosomal anomaly: 2 [PMP22_Leu16Pro, PMP22_Leu80Pro]
  Rare genetic respiratory disease: 2 [SFTPC_Leu194Pro, SFTPC_Leu188Gln]
  Ciliopathy: 2 [RHO_Cys110Tyr, RHO_Asp190Gly]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: PELIZAEUS-MERZBACHER DISEASE (4x) [PLP1_Thr147Pro, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  Disease: SURFACTANT METABOLISM DYSFUNCTION, PULMONARY, 2 (2x) [SFTPC_Leu194Pro, SFTPC_Leu188Gln]
  Disease: RETINITIS PIGMENTOSA 4 (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  Disease: SPASTIC PARAPLEGIA 2, X-LINKED (2x) [PLP1_Ile152Thr, PLP1_Leu189Pro]
  inheritance_pattern: autosomal dominant (4x) [PMP22_Leu16Pro, SFTPC_Leu194Pro, SFTPC_Leu188Gln, PMP22_Leu80Pro]
  inheritance_pattern: x-linked recessive (4x) [PLP1_Thr147Pro, PLP1_Ile152Thr, PLP1_Thr43Ile, PLP1_Leu189Pro]
  inheritance_pattern: unknown (2x) [PLP1_Gly186Cys, PLP1_Cys193Tyr]
  inheritance_pattern: autosomal recessive (2x) [RHO_Cys110Tyr, RHO_Asp190Gly]
  pLDDT_Category: >90 (9x) [PMP22_Leu16Pro, PLP1_Thr147Pro, SFTPC_Leu194Pro, SFTPC_Leu188Gln, RHO_Cys110Tyr, PLP1_Thr43Ile, PLP1_Cys193Tyr, RHO_Asp190Gly, PMP22_Leu80Pro]
  pLDDT_Category: 70-90 (3x) [PLP1_Gly186Cys, PLP1_Ile152Thr, PLP1_Leu189Pro]
  RSA_Category: 10-90 (6x) [PMP22_Leu16Pro, PLP1_Thr147Pro, SFTPC_Leu194Pro, SFTPC_Leu188Gln, PLP1_Thr43Ile, PMP22_Leu80Pro]
  RSA_Category: <10 (6x) [PLP1_Gly186Cys, RHO_Cys110Tyr, PLP1_Ile152Thr, PLP1_Cys193Tyr, RHO_Asp190Gly, PLP1_Leu189Pro]
  Transmembrane (UniProt): Helical (2x) [PMP22_Leu16Pro, PMP22_Leu80Pro]
  Topological domain (UniProt): Extracellular (3x) [RHO_Cys110Tyr, PLP1_Thr43Ile, RHO_Asp190Gly]
```







Cluster 5: Comprehensive Analysis

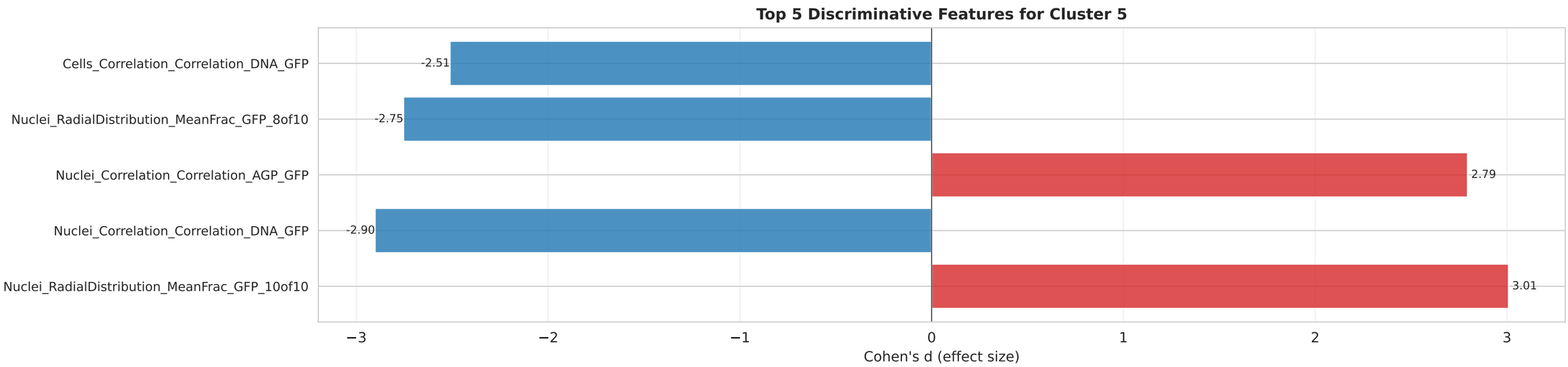
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CLUSTER 5 ANALYSIS
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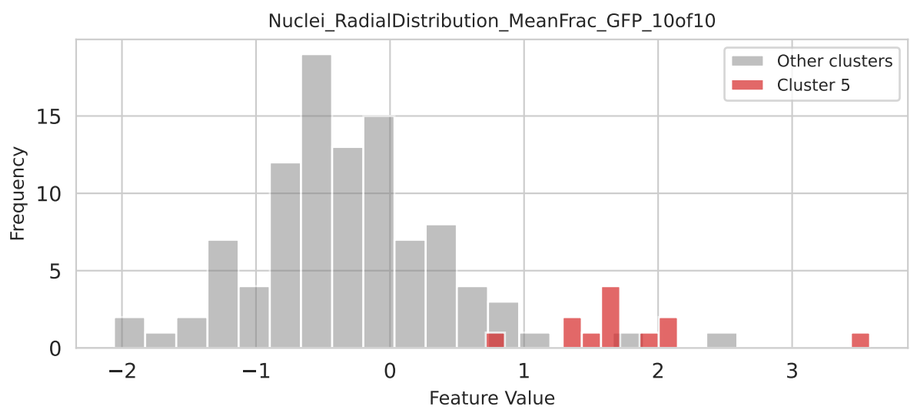
Size: 12 variants
Mean score: 0.991
Hit variants: 12 (100.0%)

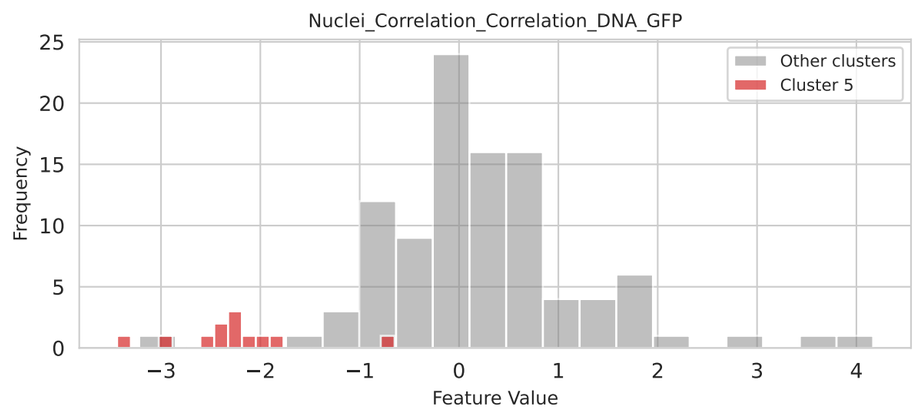
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AIPL1_Met79Thr          |  MLH1_Ile36Asn          |  MLH1_Cys39Tyr          |  MLH1_Gly147Arg
  MLH1_Val185Gly          |  MLH1_Gln701His         |  NR0B1_Arg267Pro        |  NR0B1_Tyr380Asp
  NR0B1_Arg425Ile         |  NR0B1_Asn440Ile        |  SMAD3_Trp94Leu         |  SMAD3_Cys109Tyr

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic immune disease: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Rare genetic tumor: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Inherited cancer-predisposing syndrome: 5 [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Ile36Asn, MLH1_Cys39Tyr]
  Rare genetic gynecological and obstetrical diseases: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic cardiac disease: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic developmental defect during embryogenesis: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic urogenital disease: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic endocrine disease: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Genetic infertility: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Rare genetic skin disease: 4 [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (4x) [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, MLH1_Cys39Tyr]
  Disease: ADRENAL HYPOPLASIA, CONGENITAL (4x) [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  Disease: LYNCH SYNDROME 1 (2x) [MLH1_Gln701His, MLH1_Cys39Tyr]
  inheritance_pattern: autosomal dominant (5x) [MLH1_Gln701His, MLH1_Val185Gly, MLH1_Gly147Arg, SMAD3_Trp94Leu, MLH1_Cys39Tyr]
  inheritance_pattern: x-linked recessive (4x) [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, NR0B1_Asn440Ile, NR0B1_Arg267Pro]
  inheritance_pattern: unknown (3x) [SMAD3_Cys109Tyr, MLH1_Ile36Asn, AIPL1_Met79Thr]
  pLDDT_Category: >90 (9x) [NR0B1_Arg425Ile, MLH1_Val185Gly, NR0B1_Asn440Ile, MLH1_Gly147Arg, SMAD3_Trp94Leu, SMAD3_Cys109Tyr, MLH1_Ile36Asn, MLH1_Cys39Tyr, AIPL1_Met79Thr]
  pLDDT_Category: 70-90 (2x) [NR0B1_Tyr380Asp, NR0B1_Arg267Pro]
  RSA_Category: <10 (8x) [NR0B1_Arg425Ile, NR0B1_Tyr380Asp, MLH1_Val185Gly, MLH1_Gly147Arg, SMAD3_Cys109Tyr, MLH1_Ile36Asn, MLH1_Cys39Tyr, AIPL1_Met79Thr]
  RSA_Category: 10-90 (3x) [NR0B1_Asn440Ile, NR0B1_Arg267Pro, SMAD3_Trp94Leu]
```







Cluster 6: Comprehensive Analysis

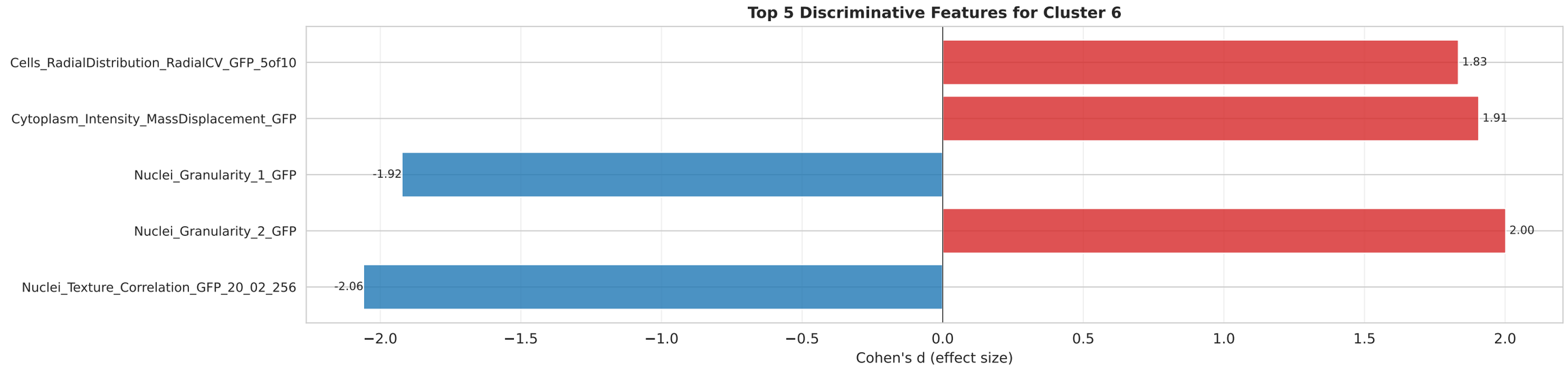
```
CLUSTER 6 ANALYSIS
================================================================================================================

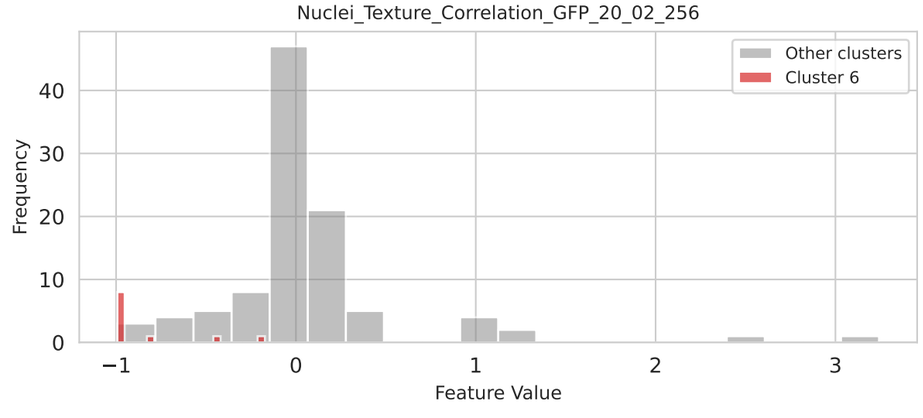
Size: 11 variants
Mean score: 0.937
Hit variants: 11 (100.0%)

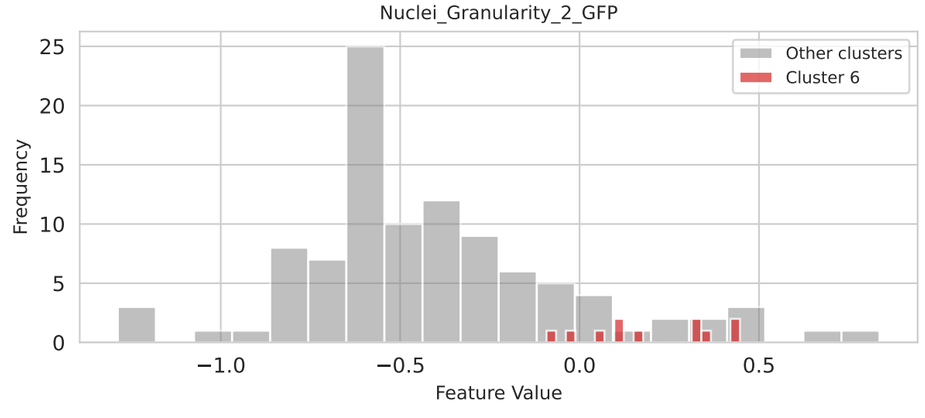
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  MLH1_His318Gln          |  MLH1_Ser508Arg          |  MLH1_Lys618Arg          |  MLH1_Pro640Ser
  MLH1_Lys670Arg          |  MLH1_Cys680Arg          |  MLH1_Ala681Thr          |  MLH1_Val716Met
  MSRB3_Cys82Gly          |  RAD51D_Met308Val        |  ZC4H2_Leu66His

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic tumor: 9 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, RAD51D_Met308Val, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Rare genetic immune disease: 8 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Inherited cancer-predisposing syndrome: 8 [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  Rare genetic developmental defect during embryogenesis: 1 [ZC4H2_Leu66His]
  Rare genetic neurological disorder: 1 [ZC4H2_Leu66His]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  Disease: LYNCH SYNDROME 2 (7x) [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, MLH1_His318Gln, MLH1_Ser508Arg]
  inheritance_pattern: autosomal dominant (7x) [MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met, MLH1_Lys618Arg, RAD51D_Met308Val, MLH1_His318Gln, MLH1_Ser508Arg]
  pLDDT_Category: >90 (9x) [MLH1_Ala681Thr, MSRB3_Cys82Gly, ZC4H2_Leu66His, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Lys618Arg, RAD51D_Met308Val, MLH1_Lys670Arg, MLH1_Ser508Arg]
  pLDDT_Category: 70-90 (2x) [MLH1_Val716Met, MLH1_His318Gln]
  RSA_Category: 10-90 (7x) [MSRB3_Cys82Gly, ZC4H2_Leu66His, MLH1_Lys618Arg, RAD51D_Met308Val, MLH1_His318Gln, MLH1_Lys670Arg, MLH1_Ser508Arg]
  RSA_Category: <10 (4x) [MLH1_Ala681Thr, MLH1_Cys680Arg, MLH1_Pro640Ser, MLH1_Val716Met]
```







Cluster 7: Comprehensive Analysis

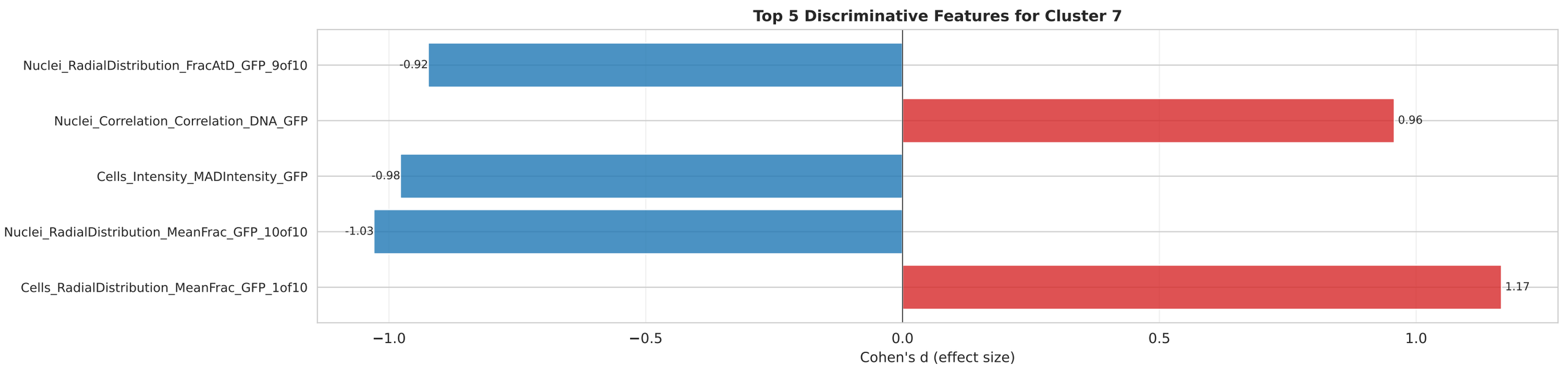
```
CLUSTER 7 ANALYSIS
================================================================================================================

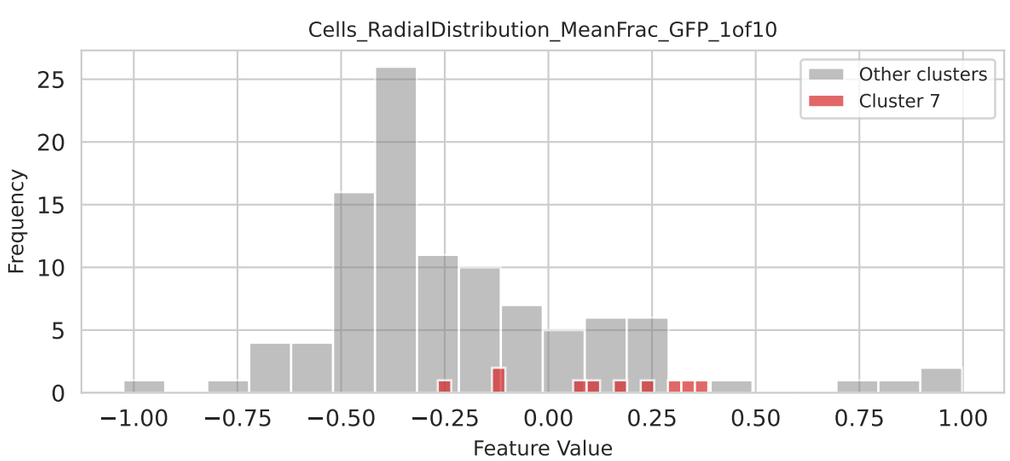
Size: 10 variants
Mean score: 0.955
Hit variants: 10 (100.0%)

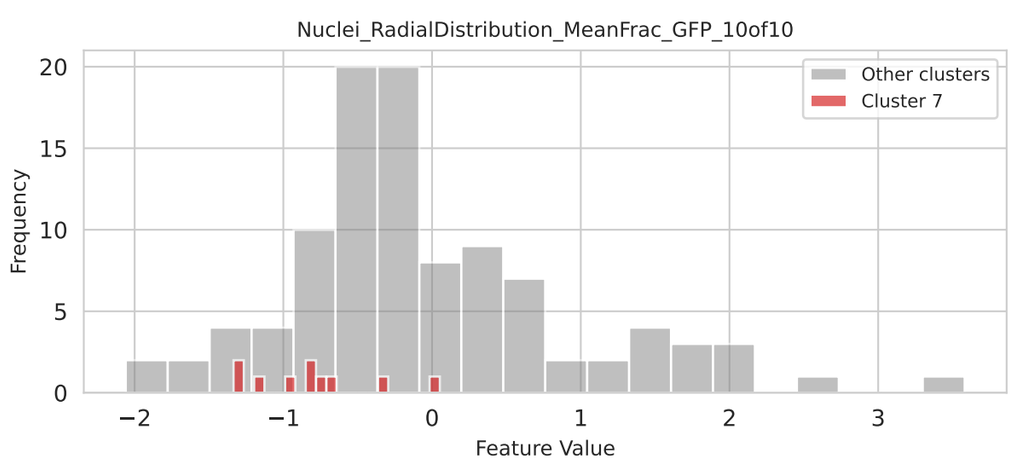
ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  AGXT_Ser218Leu          |  DCX_Arg102Cys          |  DCX_Pro191Arg          |  HPRT1_Asp201Gly
  KRT6A_Ile178Asn         |  RHO_Pro267Thr          |  SMAD3_Glu239Lys        |  SMAD3_Met255Arg
  SMAD3_Val410Phe         |  TAT_Pro220Ser

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic developmental defect during embryogenesis: 3 [DCX_Arg102Cys, DCX_Pro191Arg, KRT6A_Ile178Asn]
  Rare genetic neurological disorder: 3 [DCX_Arg102Cys, DCX_Pro191Arg, HPRT1_Asp201Gly]
  Rare genetic eye disease: 3 [RHO_Pro267Thr, AGXT_Ser218Leu, TAT_Pro220Ser]
  Rare genetic vascular disease: 3 [SMAD3_Glu239Lys, SMAD3_Val410Phe, SMAD3_Met255Arg]
  Rare genetic systemic or rheumatologic disease: 3 [SMAD3_Glu239Lys, SMAD3_Val410Phe, SMAD3_Met255Arg]
  Rare inborn errors of metabolism: 3 [AGXT_Ser218Leu, TAT_Pro220Ser, HPRT1_Asp201Gly]
  Rare genetic renal disease: 2 [AGXT_Ser218Leu, HPRT1_Asp201Gly]
  Rare genetic skin disease: 2 [KRT6A_Ile178Asn, TAT_Pro220Ser]
  Ciliopathy: 1 [RHO_Pro267Thr]
  Rare genetic hematologic disease: 1 [HPRT1_Asp201Gly]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  inheritance_pattern: unknown (7x) [DCX_Arg102Cys, DCX_Pro191Arg, RHO_Pro267Thr, SMAD3_Val410Phe, SMAD3_Met255Arg, KRT6A_Ile178Asn, TAT_Pro220Ser]
  pLDDT_Category: >90 (7x) [DCX_Arg102Cys, RHO_Pro267Thr, SMAD3_Glu239Lys, AGXT_Ser218Leu, SMAD3_Val410Phe, SMAD3_Met255Arg, HPRT1_Asp201Gly]
  RSA_Category: 10-90 (4x) [DCX_Arg102Cys, RHO_Pro267Thr, KRT6A_Ile178Asn, HPRT1_Asp201Gly]
  RSA_Category: <10 (4x) [SMAD3_Glu239Lys, AGXT_Ser218Leu, SMAD3_Val410Phe, SMAD3_Met255Arg]
```







Cluster 8: Comprehensive Analysis

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CLUSTER 8 ANALYSIS
================================================================================================

Size: 10 variants
Mean score: 0.957
Hit variants: 10 (100.0%)

ALL VARIANTS (sorted by gene, position):
--------------------------------------------------
  APOA1_Val180Glu         |  COMP_Asp319Val          |  IL10_Gly15Arg           |  IMPDH1_Arg154Trp
  IMPDH1_Asp275Asn        |  MLH1_Ser44Phe           |  SFTPA2_Phe198Ser        |  SFTPC_Ile73Thr
  SHOC2_Ser2Gly           |  SOS2_Val869Ile

DISEASE MODULES (detailed):
--------------------------------------------------
  Rare genetic eye disease: 3 [SHOC2_Ser2Gly, IMPDH1_Asp275Asn, IMPDH1_Arg154Trp]
  Rare genetic immune disease: 2 [MLH1_Ser44Phe, IL10_Gly15Arg]
  Rare genetic developmental defect during embryogenesis: 2 [SHOC2_Ser2Gly, COMP_Asp319Val]
  Ciliopathy: 2 [IMPDH1_Asp275Asn, IMPDH1_Arg154Trp]
  Rare genetic respiratory disease: 1 [SFTPC_Ile73Thr]
  Rare genetic systemic or rheumatologic disease: 1 [APOA1_Val180Glu]
  Rare genetic renal disease: 1 [APOA1_Val180Glu]
  Rare genetic endocrine disease: 1 [APOA1_Val180Glu]
  Rare inborn errors of metabolism: 1 [APOA1_Val180Glu]
  Rare genetic tumor: 1 [MLH1_Ser44Phe]

ANNOTATION ENRICHMENTS (values appearing ≥2 times):
--------------------------------------------------
  inheritance_pattern: autosomal dominant (6x) [SFTPC_Ile73Thr, MLH1_Ser44Phe, SFTPA2_Phe198Ser, IMPDH1_Asp275Asn, SOS2_Val869Ile, IMPDH1_Arg154Trp]
  inheritance_pattern: unknown (2x) [IL10_Gly15Arg, COMP_Asp319Val]
  pLDDT_Category: >90 (4x) [MLH1_Ser44Phe, SFTPA2_Phe198Ser, SOS2_Val869Ile, COMP_Asp319Val]
  pLDDT_Category: ⩽50 (2x) [SHOC2_Ser2Gly, IL10_Gly15Arg]
  RSA_Category: 10-90 (4x) [SFTPC_Ile73Thr, APOA1_Val180Glu, SHOC2_Ser2Gly, COMP_Asp319Val]
  RSA_Category: <10 (3x) [MLH1_Ser44Phe, SFTPA2_Phe198Ser, SOS2_Val869Ile]
```

